Durability Prediction for GFRP Reinforcing Bars Using Short-Term Data of Accelerated Aging Tests

Yi Chen[1]; Julio F. Davalos[2]; and Indrajit Ray[3]

Abstract: This paper presents a procedure based on the Arrhenius relation to predict the long-term behavior of glass fiber-reinforced polymer (GFRP) bars in concrete structures, based on short-term data from accelerated aging tests. GFRP reinforcing bars were exposed to simulated concrete pore solutions at 20, 40, and 60°C. The tensile strengths of the bars determined before and after exposure were considered a measure of the durability performance of the specimens. Based on the short-term data, a detailed procedure is developed and verified to predict the long-term durability performance of GFRP bars. A modified Arrhenius analysis is included in the procedure to evaluate the validity of accelerated aging tests before the prediction is made. The accelerated test and prediction procedure used in this study can be a reliable method to evaluate the durability performance of FRP composites exposed to solutions or in contact with concrete.

DOI: 10.1061/(ASCE)1090-0268(2006)10:4(279)

CE Database subject headings: Durability; Aging; Fiber reinforced polymers; Glass fiber; Predictions; Reinforcement; Bars.

Introduction

Corrosion of steel reinforcement is a major cause of deterioration of reinforced concrete structures. Fiber-reinforced polymer (FRP) reinforcing bars are being increasingly used in concrete structures due to their resistance to corrosion, high strength-to-weight ratio, good fatigue properties, and ease of handling (Uomoto et al. 2002). But among the various FRP products, glass fiber-reinforced polymer (GFRP) bars, which are extensively used in civil engineering structures due to their significantly low cost, are readily attacked by concrete pore solutions having pH values of 12.4 to 13.7 (Diamond 1981; Anderson et al. 1989; Bank and Gentry 1995).

Glass fibers are known to be susceptible to attack by water and acidic and alkaline solutions (Bank and Gentry 1995). The most severe degradation is observed in alkaline solutions. The main attack mechanisms include etching, leaching, and embrittlement. The matrices of FRP bars are intended to protect the fibers from harmful agents, but hydrolysis, plasticization, and swelling due to alkaline solution may lead to degradation of the matrix itself (Karbhari and Chu 2005). The fiber/matrix interphase, an inhomogeneous region, may also degrade easily due to matrix osmotic cracking and interfacial debonding and delamination (Bradshaw and Brinson 1997).

Though numerous studies have been conducted on the durability of composites used for aerospace or naval applications, long-term performance data for GFRP bars in civil engineering applications are still not available (Micelli and Nanni 2004). Design guidelines for FRP-reinforced concrete structures recommend strength reduction factors to account for environmental effects and sustained stress (ACI 2001; JSCE 1997). However, these factors are offered without support by significant experimental data (Greenwood 2002; Nkurunziza et al. 2005). This necessitates development of critical test methods by which long-term durability performance of GFRP bars in a concrete environment can be assessed, in order to establish a 50- to 75-year life-cycle performance of GFRP-reinforced structures.

To obtain durability results of GFRP reinforcing bars within a reasonable time, accelerated test methods have generally been adopted by researchers by immersing bare GFRP bars in simulated solutions at elevated temperatures. Since degradation of GFRP bars in solutions mainly depends on diffusion and chemical reactions (Micelli and Nanni 2004), both of these factors can be accelerated by raising temperatures; thus elevated temperatures were used as an accelerating factor by several researchers (Micelli and Nanni 2004; Porter and Barnes 1998; Dejke 2001). Sustained loading during environmental exposure was also used by other researchers such as Benmokrane et al. (2002) and Sen et al. (2002), while a combination of elevated temperatures and sustained loading was used as an accelerating factor by Rahman et al. (1998) and Nkurunziza et al. (2003).

In recent studies (Mukherjee and Arwikar 2005, Almusallam and Al-Salloum 2006), GFRP bars were embedded in concrete beams under sustained load in solutions at elevated temperatures. ASTM E 632 (ASTM 2000) provided a general approach for conducting accelerated tests of construction materials. Recently, ACI Committee 440 (ACI 2004) proposed an accelerated test

[1]Doctoral Student and Research Assistant, Dept. of Civil and Environmental Engineering, West Virginia Univ., Morgantown, WV 26506-6103. E-mail: sjtuchenyi@yahoo.com

[2]Benedum Distinguished Teaching Professor, Dept. of Civil and Environmental Engineering, West Virginia Univ., Morgantown, WV 26506-6103 (corresponding author). E-mail: Julio.Davalos@mail.wvu.edu

[3]Research Assistant Professor, Dept. of Civil and Environmental Engineering, West Virginia Univ., Morgantown, WV 26506-6103.

Note. Discussion open until January 1, 2007. Separate discussions must be submitted for individual papers. To extend the closing date by one month, a written request must be filed with the ASCE Managing Editor. The manuscript for this paper was submitted for review and possible publication on August 29, 2005; approved on November 16, 2005. This paper is part of the ***Journal of Composites for Construction***, Vol. 10, No. 4, August 1, 2006. ©ASCE, ISSN 1090-0268/2006/4-279–286/$25.00.

J. Compos. Constr. 2006.10:279-286.

method for evaluating alkali resistance of FRP bars. The test consists of subjecting FRP bars to alkali environments with or without stress at a temperature of 60°C. But questions still remain on procedures, magnitude of sustained loads to be applied to the specimens, types of simulated alkaline environment to be used, and duration of exposures.

Following accelerated tests, analyses were often carried out on the short-term data, and predictions for long-term behavior were made based on various models. Following the procedure used by Litherland et al. (1981), prediction equations were suggested for GFRP bars by Porter and Barnes (1998), but the approach successfully used in GRC (glass fiber-reinforced concrete) in Litherland et al. (1981) may not be applicable to GFRP bars, since the fibers are embedded in polymers.

Recently, Caceres et al. (2000) adopted the time-temperature superposition method to analyze the short-term data from accelerated testing of FRP composites used in naval waterfront infrastructures, but details of how to superimpose the accelerated data sets were not provided. Dejke (2001) suggested a time shift factor based on the Arrhenius relation to correlate strength retentions of concrete-encased GFRP bars exposed to solutions at 20 and 60°C. Since data from only two different exposure temperatures were used to calculate the activation energy, the time shift factor may not be generally applicable. Bank et al. (2003) provided a protocol for predicting long-term property values of FRP composites subjected to accelerated aging based on the Arrhenius relation. However, a recent study by Gonenc (2003) showed that this method cannot be directly applied to GFRP bars.

Diffusion models based on Fick's law were also used to correlate the losses in tensile strength with the moisture absorption of FRP bars by Katsuki and Uomoto (1995) and Tannous and Saadatmanesh (1998). But the main assumption of this approach, that the matrix and fibers within the depth of the damaged zone (the area with diffused chemicals) were completely degraded, may not be accurate, as shown by Aguíñiga et al. (2004). This approach cannot be applied to specimens exposed to water since the concentration of solution is required in the analysis. Also, high-quality matrices typically used currently may not allow diffusion of ions in measurable quantities (Dejke 2001).

From the literature review, one can conclude that no consensus was reached on using a particular accelerated test method or prediction model. Also, no correlation between accelerated test results and real-life aging was suggested by the relevant ACI committee report (ACI 2004). Prediction models based on the Arrhenius relation have often been used in the published literature, such as Litherland et al. (1981) and Phani and Bose (1987), but concern remains about the implicit assumption that the elevated temperature will only increase the rate of degradation without influencing the degradation mechanism or introducing other degradation mechanisms for FRP bars.

A comprehensive study on the durability of FRP bars subjected to environmental agents such as water, seawater, alkaline solutions, wetting and drying cycles, and freezing and thawing cycles was recently reported by Chen et al. (2006). Based on that study, the most significant degradation was found to occur for GFRP bars exposed to alkaline solutions. In the present study, critical short-term data on the durability of GFRP bars are obtained through accelerated aging tests by exposing specimens to simulated concrete pore solutions at different temperatures. Based on these short-term data, a detailed procedure is developed and verified to predict the long-term behavior of GFRP bars in simulated concrete pore solutions. This procedure includes a modified Arrhenius analysis to evaluate the validity of accelerated tests before the prediction is made.

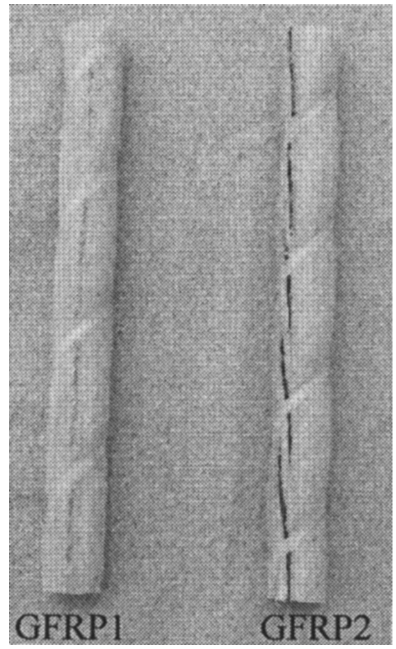


Fig. 1. GFRP bars

Experimental Program

Materials

In this study, GFRP bars made of E-glass fibers and vinyl ester resin were selected due to their wide application in civil engineering infrastructure. For the purpose of developing and verifying the proposed prediction procedure, two types of GFRP bars with a 9.53 mm diameter (No. 3) were used. The bars were produced using the same resin but different types of E-glass fibers. Both GFRP1 and GFRP2 bars were helically wrapped and slightly sand coated (Fig. 1) and were manufactured by a commercial pultrusion process. The glass fiber contents of both GFRP bars were above 70% by weight. Both types of bars were provided by the same manufacturer, and GFRP2 bar is a commercially available product.

Testing Plan

Tensile strengths of specimens were measured before and after the exposures, since changes in tensile strength values can reliably be used as indicators of the durability performance. The tensile test was conducted in accordance with ASTM D 3916-94 (ASTM 2000) with some modifications. The overall length of the specimens was 1.02 m. The end grip used in this study was a 200-mm long steel pipe cut lengthwise into two halves. After being coated with epoxy in their inner surface, the split pipes were bonded on each end of the bar. The length-to-diameter ratio of the test specimens was 64. The tests were carried out on a Baldwin machine, and the duration of the loading was 2 to 4 min for each specimen.

J. Compos. Constr. 2006.10:279-286.

Table 1. Compositions of Simulated Concrete Pore Solutions

	Quantities (g/L)		
Solution type	NaOH	KOH	$Ca(OH)_2$
1	2.4	19.6	2
2	0.6	1.4	0.037

In this study, groups of GFRP bars were tested before and after immersion in simulated pore solutions at different temperatures for different exposure times. Two types of alkaline solutions were made to simulate the pore solutions of normal concrete (NC) and high-performance concrete (HPC), respectively. Due to the increasing use of HPC in civil infrastructure, a simulated HPC pore solution was included in this study. Solutions 1 and 2 were both made with combined sodium hydroxide (NaOH), potassium hydroxide (KOH), and calcium hydroxide [$Ca(OH)_2$], according to Shi et al. (1998) and Gowripalan and Mohamed (1998). Solution 1, with a pH value of about 13.6, was intended to simulate the pore solution of NC, and Solution 2, with a pH value of about 12.7 and different proportions of hydroxides compared to Solution 1, was made to simulate the pore solution of HPC. The compositions of Solutions 1 and 2 are listed in Table 1.

Elevated temperatures of 40 and 60°C were adopted to accelerate the attack of simulated environments on specimens. These temperatures were well below the glass transition temperature of selected GFRP bars. Temperature-controlled tanks (Fig. 2) were custom designed for specimens exposed to solutions at temperatures of 40 and 60°C. For exposure of specimens in solutions at room temperature (RT, about 20°C), polypropylene heavy-wall tanks were used.

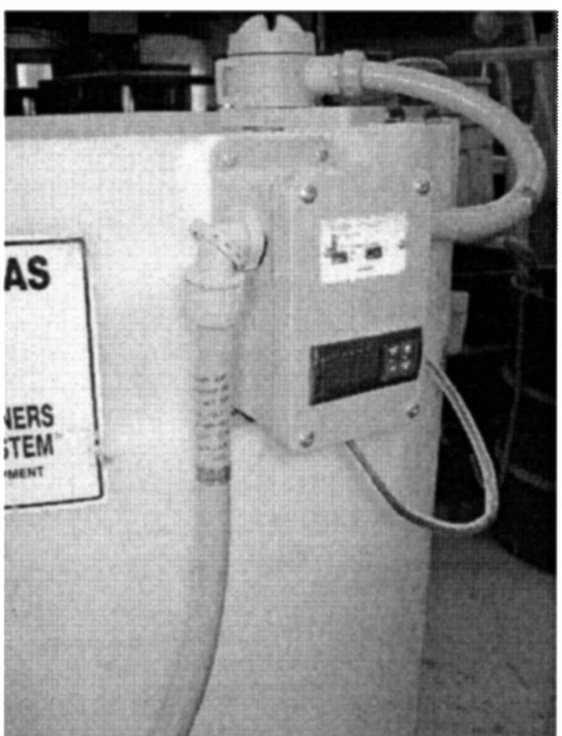
For illustrative purpose to develop and verify the proposed prediction procedure, GFRP1 bars were immersed in Solution 1 and GFRP2 bars in Solution 2. The justification for exposing a particular type of GFRP bars in a specific type of solution is based on previous findings by Chen et al. (2006) of a comprehensive study on interactions of GFRP bars and various simulated solutions. In that study, it was observed that in terms of durability, GFRP2 bars performed better than GFRP1 bars, and in terms of its reactivity with GFRP bars, Solution 2 was less aggressive than Solution 1.

Since the objective of the current paper is to develop a procedure to evaluate the durability of GFRP bars, it is necessary to verify the validity of the procedure through at least two distinct conditions in terms of the reactivity between a fiber-resin system of GFRP bars and solutions. Considering this objective, two distinct conditions are formulated, in which GFRP1 bars exposed in Solution 1 are regarded as a high-reactivity condition, and GFRP2 bars exposed in Solution 2 are considered as a relatively low-reactivity condition. The applicability of the model when including these two distinct conditions will enable evaluation of its accuracy for predicting durability from critical short-term data of accelerated tests.

The environmental exposure plan for specimens is summarized in Table 2. After each exposure, usually three to six GFRP bars were removed from solutions and tested for tensile strength retention compared to baseline specimens.

Test Results and Discussion

During the tensile test of dry specimens, both types of GFRP bars showed an approximately linear behavior up to failure and failed through the rupture of fibers. The failure of both types of GFRP bars was accompanied by the separation of fibers and the rupture of spiral bundles on the deformed surface of bars, as shown in


(a)

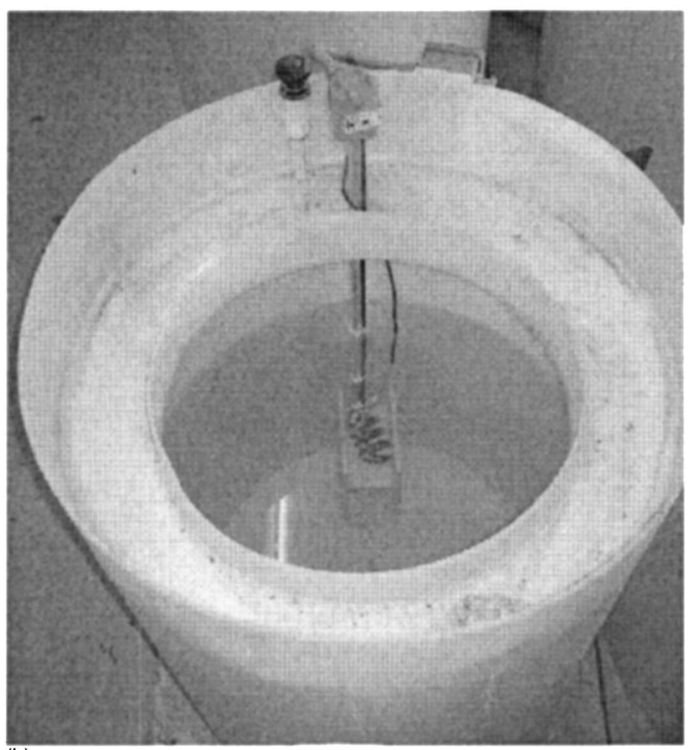

(b)

Fig. 2. Temperature-controlled tank: (a) digital temperature control; (b) Teflon-coated immersion heater

J. Compos. Constr. 2006.10:279-286.

Table 2. Environmental Exposure Plan of GFRP Bars

Bar type	Solution type	Temperature (°C)	Exposure time (days)	(days)	(days)	(days)
GFRP1	1	60	60	90	120	240
		40	60	90	120	240
		20	60	90	120	240
GFRP2	2	60	60	70	90	120
		40	60	70	90	120
		20	60	70	90	120

Fig. 3. Similar but less catastrophic failure was observed for conditioned specimens, and similar tensile failure modes of GFRP bars were also observed by Micelli and Nanni (2004).

The tensile test results of unconditioned and conditioned specimens are summarized in Figs. 4–6 (the overbar represents a standard deviation). Each data represents an average of three to six test results. As shown in Figs. 4(a and b) the tensile strengths for pristine GFRP1 and GFRP2 bars were 925 and 771 MPa, respectively. Note that the tensile strength of GFRP1 bars reduced to 352 MPa after only 120 days exposure to Solution 1 at 60°C, and the tensile strength of GFRP 2 bars reduced to 455 after 120 days exposure to Solution 2 at 60°C.

Figs. 5 and 6 show that the tensile strength decreased with an increase in exposure time for both GFRP bars at all temperatures, and degradation was more severe for specimens in solutions at higher temperatures. A comparison of GFRP bar and solution interaction confirms the previous finding of greater reactivity of the GFRP1–Solution 1 combination. One can also observe that the degradation was significant for both GFRP bars in such short exposure times compared to their expected service life. The rapid degradation may be due to the direct exposure of specimens to strong alkaline solutions. The degradation of the bars was made visibly apparent by white blisters on the surface, as described in Chen et al. (2006). The blisters may be the manifestation of osmosis in the surface of bars.

Predictions of Long-Term Behavior of GFRP Bars

The basis for the proposed prediction is the Arrhenius relation, which is introduced first. Then a detailed procedure is carried out on the short-term data to predict the long-term behavior of GFRP bars. A modified Arrhenius analysis is included in the procedure to determine the validity of the accelerated test. Then, based on the results of this study, comments and discussions are given on both the present prediction procedure and those proposed by other researchers.

Arrhenius Relation

In the Arrhenius relation, the degradation rate is expressed as follows (Nelson 1990):

$$k = A\exp\left(\frac{-E_a}{RT}\right) \tag{1}$$

where k=degradation rate (1/time); A=constant of the material and degradation process; E_a=activation energy; R=universal gas constant; and T=Kelvin temperature. The primary assumption of this model is that the single dominant degradation mechanism of the material will not change with time and temperature during the exposure, but the rate of degradation will be accelerated with the increase in temperature. Eq. (1) can be transformed into

$$\frac{1}{k} = \frac{1}{A}\exp\left(\frac{E_a}{RT}\right) \tag{2}$$

$$\ln\left(\frac{1}{k}\right) = \frac{E_a}{R}\frac{1}{T} - \ln(A) \tag{3}$$

From Eq. (2), the degradation rate k can be expressed as the inverse of time needed for a material property to reach a given value. From Eq. (3) one can further observe that the logarithm of time needed for a material property to reach a given value is a linear function of $1/T$ with the slope of E_a/R.

Prediction Procedure

For the first step, the relationship between tensile strength retention (the percentage of residual strength over original tensile strength) of GFRP bars and exposure time for the accelerated test was defined as

$$Y = 100\exp\left(-\frac{t}{\tau}\right) \tag{4}$$

where Y=tensile strength retention value (%); t=exposure time; and $\tau=1/k$, as expressed in Eq. (2). The form of this equation was modified from a study by Phani and Bose (1987) by assuming that GFRP bars degraded completely at infinite exposure time. The data of Figs. 5 and 6 were used in Eq. (4) in order to obtain the coefficient τ by regression analysis. Corresponding τ values and correlation coefficients (r) are summarized in Table 3, with all regression lines having a correlation coefficient above 0.93. Thus, the time to reach a given tensile strength retention at different temperatures can be approximately calculated through Eq. (4).

In the second step, the Arrhenius relationships were obtained by plotting the natural log of time to reach 50, 60, 70, and 80% tensile strength of GFRP bars versus $1/T$ (the inverse of exposure temperature) in Figs. 7 and 8. Straight lines were fitted to the data

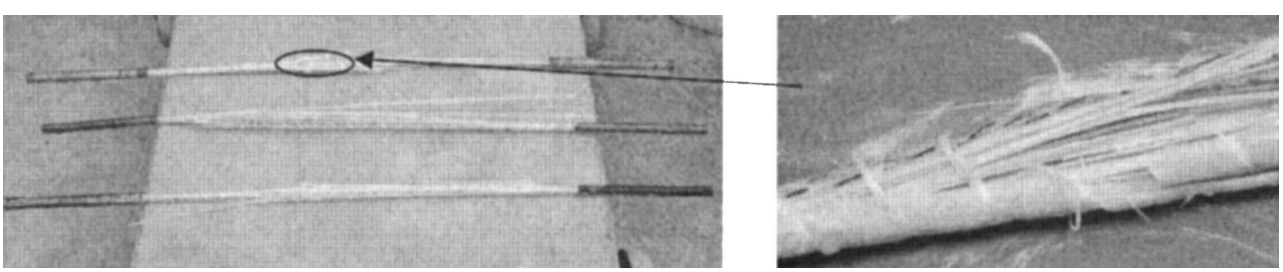


Fig. 3. Typical failure mode of GFRP bars

J. Compos. Constr. 2006.10:279-286.

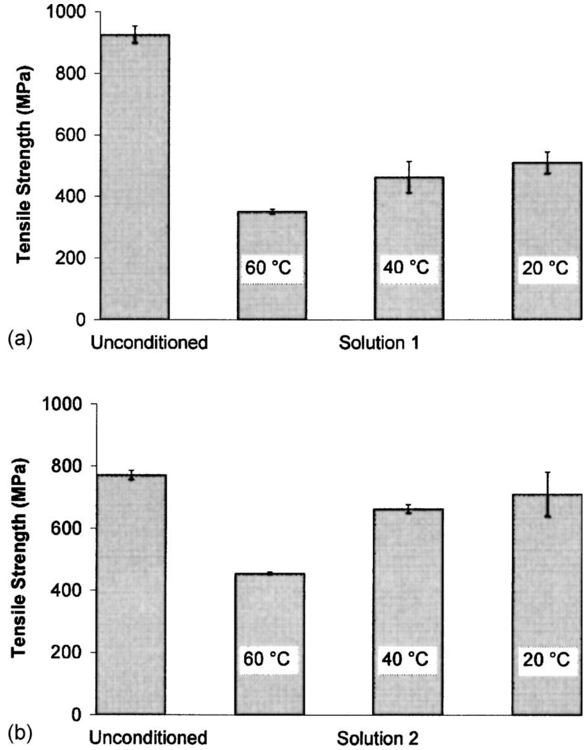


Fig. 4. Tensile strength of GFRP bars: (a) tensile strength of GFRP1 bars in Solution 1 after 120 days exposure; (b) tensile strength of GFRP2 bars in Solution 2 after 120 days exposure

with the assumption that the degradation rate was a function of temperature as expressed in Eq. (3). Eq. (2) was also fitted to the data of Table 3 to obtain E_a/R. From the analysis, the regression coefficients (E_a/R) and correlation coefficients (r) are listed in Table 4. The correlation coefficients for all regression lines were close to 1, and straight lines in Arrhenius plots for different strength retentions were nearly parallel to each other (the slopes of straight lines are E_a/R). This implies that the Arrhenius relation can be used to describe the degradation rate of GFRP bars, as the degradation mechanism may not change with temperature and time during exposure in the range tested. Moreover, Eq. (4) can be used to define the time and temperature dependence of tensile strength for GFRP bars exposed to alkaline solutions.

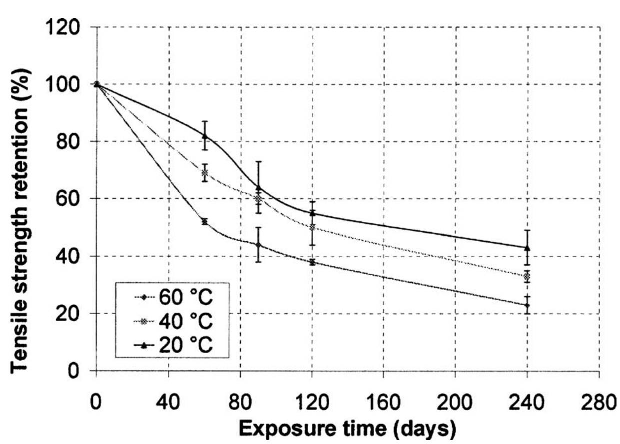


Fig. 5. Tensile strength retention of GFRP1 bars exposed to Solution 1 at 20, 40, and 60°C

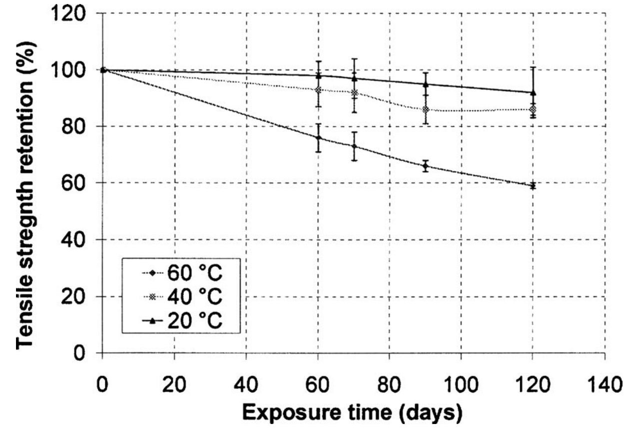


Fig. 6. Tensile strength retention of GFRP2 bars exposed to Solution 2 at 20, 40, and 60°C

For the third step, the acceleration factor (AF) for the same solution at two different temperatures can be obtained from previous Arrhenius plots. The AF can be expressed as

$$\mathrm{AF} = \frac{t_0}{t_1} = \frac{c/k_0}{c/k_1} = \frac{k_1}{k_0} = \frac{A\exp\left(\dfrac{-E_a}{RT_1}\right)}{A\exp\left(\dfrac{-E_a}{RT_0}\right)} = \exp\left[\frac{E_a}{R}\left(\frac{1}{T_0} - \frac{1}{T_1}\right)\right] \qquad (5)$$

where AF=acceleration factor; t_1 and t_0=times required for some property to reach a given value at temperatures of T_1 and T_0, respectively; c=constant; and k_1 and k_0=degradation rates at temperatures of T_1 and T_0, respectively, as expressed in Eq. (1). For example, for 70% retention of tensile strength of GFRP2 bars (as shown in Fig. 8), the AF for Solution 2 at 60°C in relation to 20°C can be calculated, AF=7.5. Since the fitted lines are parallel to each other (Figs. 7 and 8), the acceleration factor remains constant for all strength retention values, such as those shown from 50 to 80%, at each specified temperature level, either 40 or 60°C. The AF values with reference temperature T_0=20°C are listed in Table 5. Therefore, Eq. (5) can be fitted to data in Table 5 to obtain the AF values for other temperatures based on the reference temperature of 20°C.

For the fourth step, once the AF values for 60 and 40°C were obtained, Figs. 5 and 6 were transformed into Figs. 9 and 10 by multiplying exposure times at 60 and 40°C with corresponding AF values. Master curves for tensile strength retention versus exposure time at 20°C were obtained by fitting Eq. (4) to data in Figs. 9 and 10. As listed in Table 6, these data have correlation coefficients of 0.92 and 0.99, respectively. Interestingly, the values for τ correspond exactly to those at 20°C given in Table 3; this close correlation also confirms the validity of this procedure.

Finally, Eq. (4) with the values for τ listed in Table 6 could be used to predict the tensile strength retention at any exposure time.

Table 3. Coefficients of Regression Equations for GFRP Tensile Strength Retention

Temperature (°C)	GFRP1 bars in Solution 1		GFRP2 bars in Solution 2	
	τ	r	τ	r
60	143	0.93	222	0.99
40	200	0.98	714	0.96
20	256	0.96	1,667	0.94

J. Compos. Constr. 2006.10:279-286.

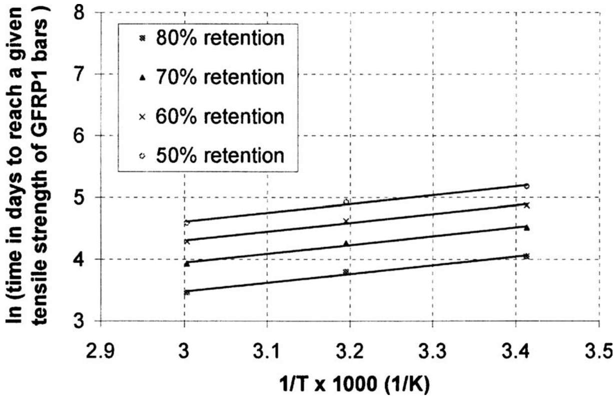


Fig. 7. Arrhenius plots of tensile strength degradation for GFRP1 bars exposed to Solution 1

For example, from the master curves, after only 178-day exposure in Solution 1 at 20°C, the tensile strength retention of GFRP1 bars will drop to 50%. For GFRP2 bars exposed to Solution 2 at 20°C, the tensile strength retention will be 50% at 1,156 days.

Discussions

The Arrhenius analysis (the first and second steps in the above prediction procedure) was also used to determine the validity of the accelerated tests. If the fitted straight lines in Arrhenius plots had a low correlation coefficient (e.g., $r<0.8$) and/or were not parallel to each other, one would infer that the degradation mechanism would have changed with temperature and time, indicating that the accelerated tests are not valid and the procedure could not be used to predict long-term behavior.

Note that the slope (E_a/R) of Arrhenius plots for GFRP1 bars is about 1,420, which was much lower than about 4,890 for GFRP2 bars, revealing that the activation energy (E_a) for degradations of GFRP1 bars in Solution 1 is much lower than for GFRP2 bars in Solution 2, resulting in different degradation rates and probably mechanisms. In a recent study (Dejke 2001), an E_a/R value of 6,300 was obtained for GFRP bars embedded in moisture concrete, and therefore the degradation of different GFRP bars in different alkaline environments will be different.

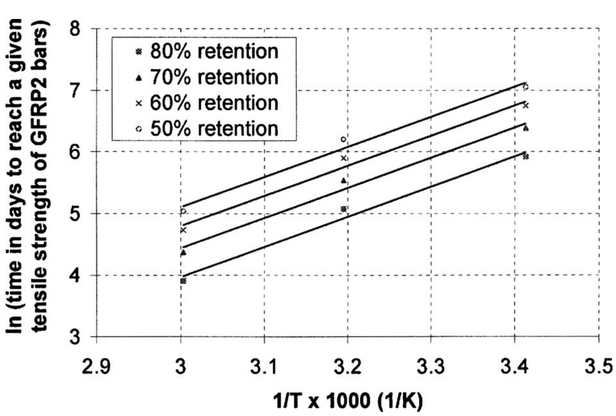


Fig. 8. Arrhenius plots of tensile strength degradation for GFRP2 bars exposed to Solution 2

Table 4. Coefficients of Regression Equations for Arrhenius Plots

Tensile strength retention (%)	GFRP1 bars in Solution 1		GFRP2 bars in Solution 2	
	E_a/R	r	E_a/R	r
50	1,420	0.99	4,891	0.99
60	1,423	0.99	4,892	0.99
70	1,420	0.99	4,891	0.99
80	1,420	0.99	4,892	0.99
Eq. (2)[a]	1,415	0.99	4,899	0.99

[a]For Eq. (2) based on data in Table 3.

So it is worth noting that if the master curve for prediction is developed for one type of GFRP bars in a specific exposure condition, it may not be applicable to other types of GFRP bars used in different environments.

From the obtained master curve, the degradation was significant for both GFRP bars in such short exposure times compared to their expected service life. Based on a diffusion model, a similar trend was also predicted for GFRP bars directly exposed to alkaline solution by Sen et al. (2002). The predicted service life for a specific E-glass/vinyl ester reinforcement used by the U.S. Navy was only between 1.6 and 4.6 years.

Prediction procedures proposed by other researchers were also used to analyze the data from this study. In the model proposed by Bank et al. (2003), the relationship between strength retention and log of exposure time was assumed linear. The Arrhenius plots were directly used to predict the service life of different strength retentions, and second group of Arrhenius plots was also obtained and used to predict the strength retentions at different exposure periods. Following the procedure proposed by Bank et al. (2003), straight lines of Arrhenius plots obtained using their approach were found not to be parallel to each other, which invalidates the procedure based on the assumption that the degradation mechanism will not change with time. The same situation was also found in a recent study (Gonenc 2001) when the procedure given in Bank et al. (2003) was applied to GFRP rods. The procedure proposed by Caceres et al. (2000) was also applied to the present data. Since no details were given on how to superimpose the accelerated test data for different temperatures, results were significantly influenced by the superposition method used. Moreover, as stated in Caceres et al. (2000), extrapolation of the master curve obtained using their procedure may lead to unrealistic predictions.

Compared to the procedures proposed by other researchers, the procedure in this study can be used for short-term data for GFRP bars successfully and efficiently. The proposed procedure can be easily carried out by defining simple plots and performing regression analysis. The results indicate that increasing the number of exposure temperatures and using longer exposure durations in accelerated tests can lead to more precise predictions. This procedure may also be used for FRP materials exposed to other

Table 5. Values for Acceleration Factors

Temperature (°C)	GFRP1 in Solution 1	GFRP2 in Solution 2
60	1.80	7.50
40	1.28	2.33
20	1.00	1.00

Note: Reference temperature is 20°C.

J. Compos. Constr. 2006.10:279-286.

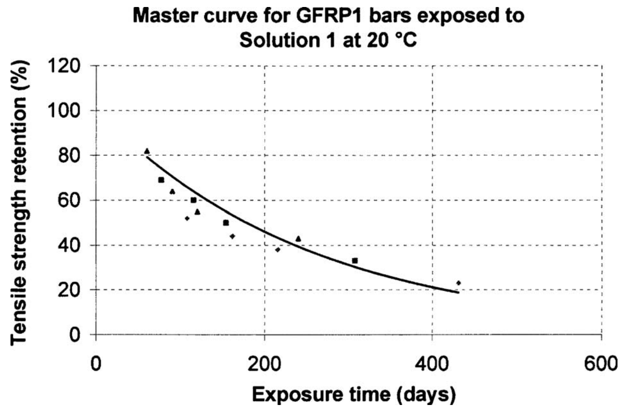


Fig. 9. Tensile strength retention versus exposure time for GFRP1 bars

solutions or even embedded in moist concrete. Due to the complexity of the degradation mechanism, no degradation model has been proposed to simulate exactly the degradation of GFRP bars in alkaline media. The accelerated test method and prediction procedure used in this study may be good options to assess the long-term durability performance of GFRP bars.

Conclusions and Recommendations

This study is part of an ongoing durability research program on FRP reinforcing bars for concrete. From the short-term data and prediction procedure presented in this paper, the following concluding remarks and recommendations can be made.

Possibly due to lower alkalinity, HPC pore solutions may be less aggressive to GFRP bars than those of NC in spite of having different ionic proportions. Simulated pore solutions with different compositions may result in different degradation mechanisms for various kinds of GFRP bars. Thus, the guidelines in ACI 440.3R2-04 (ACI 2004), in which one kind of alkaline solution is used to represent different pore solutions of various kinds of concrete, would have to be revised.

There is a dominant degradation mechanism for GFRP bars in alkaline solutions that does not appear to change with temperature or time. Elevated temperature can be used to accelerate the degradation of GFRP bars in alkaline solutions. Further, the

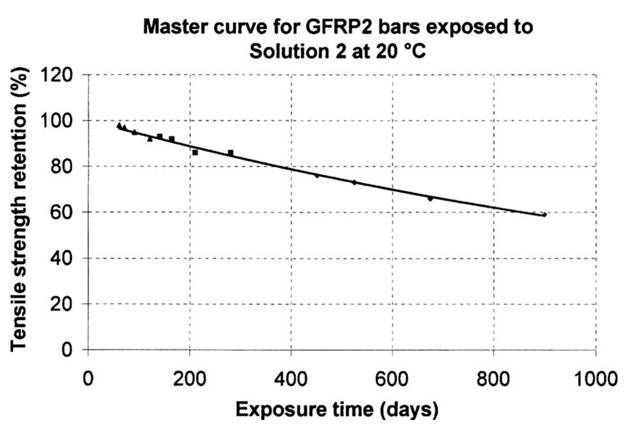


Fig. 10. Tensile strength retention versus exposure time for GFRP2 bars

Table 6. Coefficients of Regression Equations for Master Curves

Bar type	Solution type	τ	r
GFRP1	1	256	0.92
GFRP2	2	1,667	0.99

temperature dependence of the degradation rate can be described by the Arrhenius equation. Eqs. (1) and (4) can be used to predict the relation between tensile strength retention of GFRP bars with exposure time and temperature. The method presented in this study can probably be used to evaluate the durability performance of FRP composites, while the Arrhenius analysis in this procedure can be used to determine the validity of the accelerated test using elevated temperatures.

From the master curves of this study, the tensile strength retention of GFRP1 bars drops to 50% after only a half-year exposure in Solution 1 at 20°C, and for GFRP 2 bars exposed to Solution 2 at 20°C, the tensile strength retention is 50% after about 3 years. But in a recent study (Mufti et al. (2005), much less degradation was found for GFRP bars in field concrete structures. Since in the present study bare GFRP specimens were directly exposed to simulated pore solutions, the reported short-term results and the predicted tensile strength retention should be considered conservative. Note also that the master curves in this paper are only applicable to the GFRP bars tested and exposed to the specific simulated pore solutions in this study.

To investigate the durability of GFRP bars in concrete, specimens embedded in concrete should be tested. To predict long-term behaviors of GFRP bars based on the short-term data, the correlation between degradations of GFRP bars in accelerated tests and real applications needs to be investigated. Long-term data, including those in real applications, should be collected to validate accelerated tests and prediction models. Moreover, coupling effects of sustained load and environmental exposures should also be included in future studies to simulate real-life conditions.

Acknowledgments

We appreciate the financial support from the Korea Institute of Construction Technology (KICT), Republic of Korea. The funding provided by the college dean, Eugene Cilento, and department chair, David Martinelli, at West Virginia University is also gratefully acknowledged. We thank Doug Gremel of Hughes Brothers, Inc. for supplying FRP bars and for his valuable technical suggestions.

References

American Concrete Institute (ACI). (2001). "Guide for the design and construction of concrete reinforced with FRP bars." *ACI 440.1R-01*, Farmington Hills, Mich.

American Concrete Institute (ACI). (2004). "Guide test methods for fiber-reinforced polymers (FRPs) for reinforcing or strengthening concrete structures." *ACI 440.3R-04*, Farmington Hills, Mich.

Aguíñiga, F., Bradberry, T., and Trejo, D. (2004). "Time-dependent mechanical property changes of glass fiber reinforced polymers exposed to high pH environments." *Proc., 9th ASCE Aerospace Division Int. Conf. on Engineering, Construction and Operations in Challenging Environments*, ASCE, Reston, Va.

Almusallam, T. H., and Al-Salloum, Y. (2006) "Durability of GFRP rebars in concrete beams under sustained loads at severe environments." *J. Compos. Mater.*, 40(7), 623–637.

Anderson, K., Allard, B., Bengtsson, M., and Magnusson, B. (1989). "Chemical composition of cement pore solutions." *Cem. Concr. Res.*, 19(33), 327–333.

ASTM. (2000). *Annual book of ASTM standards*, ASTM, West Conshohocken, Pa.

Bank, L. C., and Gentry, T. R. (1995). "Accelerated test methods to determine the long-term behavior of FRP composite structures: Environmental effects." *J. Reinf. Plast. Compos.*, 14(6), 559–587.

Bank, C. L., Gentry, T. R., Thompson, B. P., and Russell, S. J. (2003). "A model specification for FRP composites for civil engineering structures." *Constr. Build. Mater.*, 17(6-7), 405–437.

Benmokrane, B., Wang, P., Ton-That, T. M., Rahman, H., and Robert, J.-F. (2002). "Durability of glass fiber-reinforced polymer reinforcing bars in concrete environment." *J. Compos. Constr.*, 6(3), 143–153.

Bradshaw, R. D., and Brinson, L. C. (1997) "Physical aging in polymer composites: An analysis and method for time aging time superposition." *Polym. Eng. Sci.*, 31(1), 31–34.

Caceres, A., Jamond, R. M., Hoffard, T. A., and Malvar, L. J. (2000). "Accelerated testing of fiber reinforced polymer matrix composites—Test plan." *SP-2091-SHR*, Naval Facilities Engineering Service Center, Port Hueneme, Calif.

Chen, Y., Davalos, J. F., Ray, I., and Kim, H. Y. (2005). "Accelerated aging tests for evaluations of durability performance of FRP reinforcing bars for concrete structures." *Compos. Struct.*, in press.

Dejke, V. (2001). "Durability of FRP reinforcement in concrete—Literature review and experiments." Licentiate of engineering thesis, Dept. of Building Materials, Chalmers Univ. of Technology, Goteborg, Sweden.

Diamond, S. (1981). "Effects of two Danish fly ashes on alkali contents of pore solutions of cement-fly ash pastes." *Cem. Concr. Res.*, 11(3), 383–394.

Gonenc, O. (2001). "Durability and service-life prediction of concrete reinforcing materials." MS thesis, Univ. of Wisconsin-Madison, Madison, Wis.

Gowripalan, N., and Mohamed, H. M. (1998). "Chloride-ion induced corrosion of galvanized and ordinary steel reinforcement in high-performance concrete." *Cem. Concr. Res.*, 28(8), 1119–1131.

Greenwood, M. (2002). "Creep-rupture testing to predict long-term performance." *Proc., 2nd Int. Conf. on Durability of Fiber Reinforced Polymer (FRP) Composites for Construction*, ISIS Canada, Montréal, 203–212.

Japanese Society of Civil Engineering (JSCE). (1997). "Recommendation for design and construction of concrete structures using continuous fiber reinforcing materials." *Concrete Engineering Series*, No. 23, JSCE, Tokyo (can be found at http://www.jsce.or.jp/committee/concrete/e/publication-e.html).

Karbhari, V. M., and Chu, W. (2005). "Degradation kinetics of pultruded E-glass/vinylester in alkaline media." *ACI Mater. J.*, 102(1), 34–41.

Katsuki, F., and Uomoto, T. (1995). "Prediction of deterioration of FRP rods due to alkali attack." *Non-metallic (FRP) reinforcement for concrete structures*, RILEM, London.

Litherland, K. L., Oakley, D. R., and Proctor, B. A. (1981). "The use of accelerated aging procedures to predict the long term strength of GRC composites." *Cem. Concr. Res.*, 1(11), 455–466.

Micelli, F., and Nanni, A. (2004). "Durability of FRP rods for concrete structures." *Constr. Build. Mater.*, 18(7), 491–503.

Mufti, A., et al. (2005). "Durability of GFRP reinforced concrete in field structures." *Proc., 7th Int. Symp. on Fiber Reinforced Polymer Reinforcement for Reinforced Concrete Structures*, American Concrete Institute, Detroit.

Mukherjee, A., and Arwikar, S. J. (2005). "Performance of glass fiber-reinforced polymer reinforcing bars in tropical environments. I: Structural scale tests." *ACI Struct. J.*, 102(5), 745–753.

Nelson, W. (1990). *Accelerated testing—Statistical models, test plans, and data analyses*, Wiley, New York.

Nkurunziza, G., Benmokrane, B., Debaiky, A. S., and Masmoudi, R. (2005). "Effect of sustained load and environment on long-term tensile properties of glass fiber-reinforced polymer reinforcing bars." *ACI Struct. J.*, 102(4), 615–621.

Nkurunziza, G., Cousin, P., Masmoudi, R., and Benmokrane, B. (2003). "Effect of sustained tensile stress and temperature on GFRP composite bars properties. 1: Preliminary experiment in deionised water and alkaline solution." *Int. J. Mater. Prod. Technol.*, 19(1–2), 15–27.

Phani, K. K., and Bose, N. R. (1987). "Temperature dependence of hydrothermal ageing of CSM-Laminate during water immersion." *Compos. Sci. Technol.*, 29(2), 79–87.

Porter, M. L. and Barnes, B. A. (1998). "Accelerated durability of FRP reinforcement for concrete structures." *Proc., 1st Int. Conf. on Durability of Fiber Reinforced Polymer (FRP) Composites for Construction*, ISIS Canada, Sherbrooke, Canada, 191–201.

Rahman, A. H., Lauzier, C., Kingsley, C., Richard, J., and Crimi, J. (1998). "Experimental investigation of the mechanism of deterioration of FRP reinforcement for concrete." *Proc., 2nd Int. Conf. on Composites in Infrastructure*, 501–511.

Sen, R., Mullins, G., and Salem, T. (2002). "Durability of E-glass/vinyl ester reinforcement in alkaline solution." *ACI Struct. J.*, 99(3), 369–375.

Shi, C., Stegemann, J. A., and Caldwell, R. J. (1998). "Effect of supplementary cementing materials on the specific conductivity of pore solution and its implications on the rapid chloride permeability test (AASHTO T277 and ASTM C1202) results." *ACI Mater. J.*, 95(4), 389–394.

Tannous, F. E., and Saadatmanesh, H. (1998). "Environmental effects on the mechanical properties of E-Glass FRP rebars." *ACI Mater. J.*, 95(2), 87–100.

Uomoto, T., Mutsuyoshi, H., Katsuki, F., and Misra, S. (2002). "Use of fiber reinforced polymer composites as reinforcing material for concrete." *J. Mater. Civ. Eng.*, 14(3), 191–209.

J. Compos. Constr. 2006.10:279-286.